Bi-weekly Report

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Other separators design.
2. Reactor design and sequencing with detailed kinetics.
3. One heat exchanger detailed design.
4. Simulation of overall process with recycle loops.

1 Basis

The flowsheet delivered in Report 3 was converged and its balances closed, but the reactor in it was a stoichiometric block and several items were left open at that time. Those items are listed below. The present report takes them up, and whatever is still open is given in Section 7.

Table 1: Position at the close of Report 3

Item	Report 3 status	Taken up in
Reactor model	Stoichiometric	Section 2
Vessels other than the columns	Not sized	Section 3
Heat exchangers	Shortcut duty only	Section 4
Aniline–nitrobenzene binary parameters	Databank, 83–99 °C	Section 5.1
Nitrobenzene in product, below 5 ppm	1466 ppm	Section 5.3
Aniline purity, 99.9 wt% required	99.39 wt%	Section 5.5
Aniline in wastewater, below 1 ppm	154 ppm	Section 3.5
Aniline in WATOVHD	2.515 kmol/h	Section 3.3

All simulation work is in Aspen Plus V14 with NRTL-RK as the property method. Hydrogen and nitrogen are Henry components under the HC-1 set. The process flowsheet is `che453_R4_`

FINAL.apw. The binary regression is a separate file, che453_R4_C_regress.apw, and the exchanger is designed in Aspen Exchanger Design and Rating in the file B3.EDR. The gas turbine remains a separate file on Peng–Robinson and is not linked to the process flowsheet; the purge is taken to it as a fixed feed. No costing is given in this report. Costing is Report 5.

1.1 Key results

The reactor is now a kinetic plug-flow model. It gives 99.49 per cent conversion per pass against the 99.0 per cent that was set by hand in Report 3, with selectivities of 94.06, 1.76 and 4.19 per cent to aniline, cyclohexylamine and benzene. All four recycle loops are closed and converged together, and the overall mass balance closes to 7 parts per million on 18,300 kg/h.

The product leaves at 40 °C and 1.0 bar with 99.45 wt per cent aniline against 99.39 in Report 3, and 313 ppm nitrobenzene against 1466 ppm. That reduction came from regressing the aniline–nitrobenzene pair and then adding ten stages to COL-2, which is what Report 3 said was needed. It cost no extra reboiler duty.

Total heating is 22.23 MW and the cooling water loop circulates 1691 t/h. The reactor removes a further 18.75 MW. On the thermal study, 25 mm tubes hold the bed peak at 390.7 °C against 640.4 °C for 52.5 mm tubes. The aniline recovery condenser B3 needs 132.87 m^2 and the design delivers 147.03 m^2, an over-surface of 10.7 per cent.

2 Reactor Design with Detailed Kinetics

2.1 Reaction scheme

Three reactions are taken. The main reaction makes aniline. The other two are the side reactions seen on copper catalyst, which are over-hydrogenation to cyclohexylamine and hydrodenitrogenation to benzene and ammonia.

$$\mathrm{C_6H_5NO_2 + 3\,H_2 \longrightarrow C_6H_5NH_2 + 2\,H_2O} \tag{1}$$

$$\mathrm{C_6H_5NH_2 + 3\,H_2 \longrightarrow C_6H_{11}NH_2} \tag{2}$$

$$\mathrm{C_6H_5NH_2 + H_2 \longrightarrow C_6H_6 + NH_3} \tag{3}$$

2.2 Rate expressions

Power-law kinetics are used in the vapour phase, on a molarity basis, with the rate written on reactor volume. Aspen writes the rate constant in the reference-temperature form

$$k(T) = k_0 \left(\frac{T}{T_0}\right)^{n} \exp\!\left[-\frac{E}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right] \tag{4}$$

with $T_0 = 573.15$ K and $n = 0$ for all three reactions. The parameters are in Table 1. The activation energy of the main reaction is taken from Petrov, Kumbilieva and Kirkov for an industrial copper catalyst, which gives $E_a = 35 \pm 1$ kJ/mol [3]. The two side reactions are given higher activation energies, because both of them are known to run away only at high temperature. The pre-exponential factors were fitted so that the reactor reproduces the conversion and the selectivity that were assumed in Report 3.

The half-order in hydrogen for reactions 1 and 3 represents dissociative adsorption of hydrogen on the copper surface, which is the usual assumption for this system. Reaction 2 is taken first order in

Table 2: Power-law kinetic parameters. Vapour phase, molarity basis, rate on reactor volume, $T_0 = 573.15$ K, $n = 0$.

Rxn	Reaction	k_0	E (kJ/kmol)	Concentration exponents
1	Nitrobenzene → aniline	42	35,000	NB 1, H_2 0.5
2	Aniline → cyclohexylamine	0.88	60,000	Aniline 1, H_2 1
3	Aniline → benzene + ammonia	0.41	75,000	Aniline 1, H_2 0.5

hydrogen because cyclohexylamine formation needs three molecules of hydrogen on neighbouring sites and is not limited by dissociation.

2.3 Bed geometry and pressure drop

The reactor is a multitubular fixed bed with the catalyst in the tubes and the coolant on the shell. The geometry is in Table 2.

Table 3: Reactor bed geometry.

Item	Value	Basis
Number of tubes	2200	Set by the required catalyst volume
Tube inside diameter	52.5 mm	Standard tube, see Section 2.4
Tube length	2.0 m	
Catalyst volume	9.53 m^3	
Catalyst	Cu on silica	Bulk density 1200 kg/m^3
Bed voidage	0.40	
Particle diameter	6 mm	
Gas hourly space velocity	2860 h^{-1}	On inlet volumetric flow
Superficial velocity	1.59 m/s	
Pressure drop	0.131 bar	Ergun correlation

Pressure drop goes as L/d_p^2 at a fixed superficial velocity, so the bed was set short and the particles coarse. The 2.0 m length with 6 mm particles gives 0.131 bar, which is 4.4 per cent of the operating pressure.

In the converged flowsheet this 0.131 bar is imposed as a fixed drop rather than recomputed by Ergun at every recycle iteration, which keeps the integration stable during tearing. The converged answer is the same.

2.4 Thermal behaviour and choice of tube diameter

The main reaction is strongly exothermic. The total heat released in the bed is 18.75 MW, and 8.04 MW of that, which is 43 per cent, comes off in the first 0.2 m. So the axial temperature profile is the thing that decides the tube diameter.

Two separate non-isothermal cases were run on the same catalyst volume and the same feed, with only the tube diameter changed. The results are in Table 3.

Halving the tube diameter brings the peak down by 250 °C. This is because the heat removal per unit volume goes as the surface-to-volume ratio of the tube, which is $4/d$, so the thinner tube

Table 4: Effect of tube diameter on the axial temperature profile. Same catalyst volume, same inlet condition.

Item	52.5 mm tubes	25 mm tubes
Inlet temperature	138.0 °C	138.0 °C
Peak temperature	640.4 °C	390.7 °C
Position of peak	0.341 m	0.293 m
Outlet temperature	240.4 °C	242.5 °C
Pressure drop	0.130 bar	0.279 bar

removes heat twice as fast for the same driving force. The cost is 0.15 bar of extra pressure drop, because the same gas has to pass through a narrower tube.

Copper on silica sinters well below 640 °C, so **25 mm tubes are recommended for the built unit**. The composition results are not affected by this choice, because the nitrobenzene is almost fully converted inside the first 0.5 m in both cases and the rest of the bed is only doing clean-up.

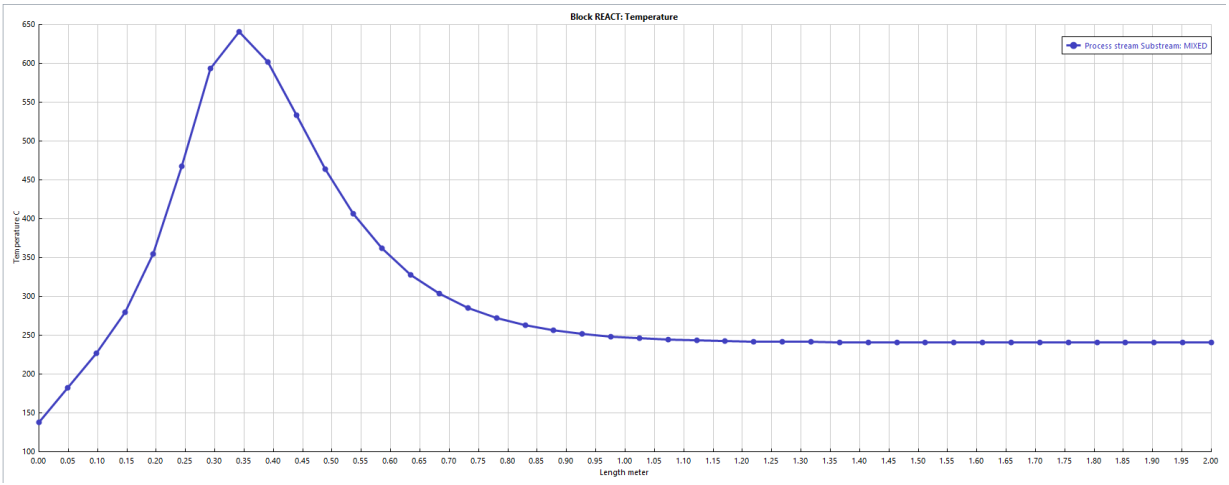


Figure 1: Axial temperature profile in the catalyst bed with 52.5 mm tubes. The peak is 640.4 °C at 0.341 m.

2.5 Reactor in the converged flowsheet

In the flowsheet the reactor is run as a constant-temperature RPlug at 300 °C. This represents a tightly cooled multitubular bed and gives the duty that the coolant has to take away, which is 18.75 MW. The non-isothermal cases in Section 2.4 are the separate thermal study on the same bed and they tell us what tube diameter holds that temperature.

Table 4 gives the reactor performance in the converged flowsheet.

The hydrogen consumption checks against stoichiometry. Three moles per mole of nitrobenzene converted gives 423.5 kmol/h, three more per mole of cyclohexylamine gives 7.4 kmol/h, and one per mole of benzene gives 5.9 kmol/h. The total is 436.8 kmol/h, which matches the simulation.

In Report 3 the stoichiometric reactor was set at 99.0 per cent conversion with selectivities of 94.95, 1.53 and 3.52 per cent. The kinetic reactor gives 99.49, 94.06, 1.76 and 4.19 per cent on the same bed. The two are close, so the rest of the flowsheet that was built on the Report 3 numbers still works on the kinetic numbers.

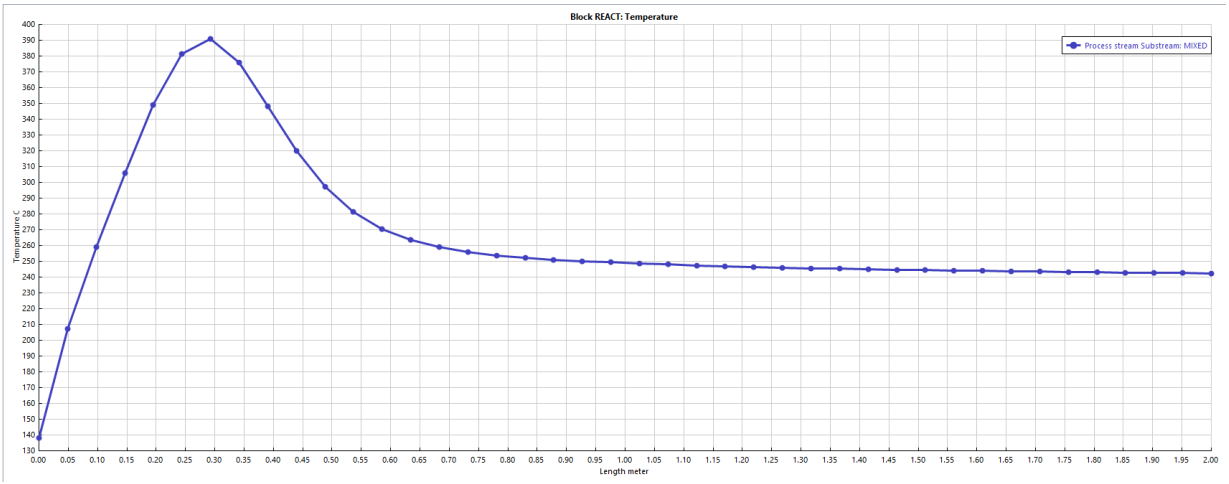


Figure 2: Axial temperature profile in the catalyst bed with 25 mm tubes. The peak is 390.7 °C at 0.293 m.

Table 5: Reactor performance in the converged flowsheet.

Item	Value
Nitrobenzene into reactor (RXFEED)	141.888 kmol/h
Nitrobenzene out (RXOUT)	0.725 kmol/h
Conversion per pass	99.49 %
Selectivity to aniline	94.06 %
Selectivity to cyclohexylamine	1.76 %
Selectivity to benzene	4.19 %
Hydrogen consumed	436.85 kmol/h
Inlet temperature	160.0 °C
Outlet temperature	300.0 °C
Heat removed	18.75 MW
Residence time	0.63 s

3 Design of the Other Separators

The distillation columns were done in Report 3. This section covers the vessels. All of them are sized on the converged flows of the final run and the checks are repeated in Section 3.6.

3.1 Flash drum

The reactor effluent is cooled and flashed to take the hydrogen-rich gas off the top for recycle. The drum is a vertical knockout drum with a demister.

The diameter comes from the Souders–Brown equation,

$$u_{\max} = K \sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \tag{5}$$

with $K = 0.1067$ m/s, which is 0.35 ft/s, the standard value for a vertical drum with a demister. The length comes from a five-minute liquid holdup plus 1.5 m of vapour space above the high

liquid level, which covers the clearance to the demister, the demister pad itself and the disengaging height above it.

Table 6: Flash drum sizing.

Item	Value	Note
Vapour flow	2.522 m^3/s	VAP-OUT at 40 °C, 2.5 bar
Vapour density	0.531 kg/m^3	
Liquid flow	17.43 m^3/h	LIQ-OUT
Liquid density	1064.9 kg/m^3	
Maximum vapour velocity $u_{\max}$	4.78 m/s	Souders–Brown
Minimum diameter	0.820 m	
Selected diameter	0.92 m	
Actual vapour velocity	3.79 m/s	79 % of $u_{\max}$
Selected length, tangent to tangent	3.8 m	$L/D = 4.1$
Liquid holdup time	5.27 min	

3.2 Decanter

The condensed reactor effluent separates into an organic phase rich in aniline and an aqueous phase. A horizontal decanter with $L/D = 5$ is used.

The dispersed phase was identified with the Selker–Sleicher criterion as given by Towler and Sinnott [6],

$$\psi = \frac{Q_L}{Q_H}\left(\frac{\rho_L \mu_H}{\rho_H \mu_L}\right)^{0.3} \tag{6}$$

which gives $\psi = 1.65$. This falls in the band where either phase can be the dispersed one, so the vessel was sized on the slower of the two settling velocities, which is the aqueous droplet rising through the organic phase.

Settling is by Stokes law on a 150 μm design droplet,

$$u = \frac{d^2 g\,\Delta\rho}{18\,\mu_c} \tag{7}$$

The settling time is below the residence time and the continuous-phase velocity along the interface is below the settling velocity, so both of the usual decanter checks are satisfied.

3.3 Second decanter on the column overhead

The overhead from the first column, stream WATOVHD, carries 2.61 kmol/h of aniline along with water and benzene. A second decanter on this stream recovers 76.4 per cent of that aniline, which is 1.99 kmol/h or 185 kg/h, into an organic phase.

It is reported open loop, the recovered organic not being fed back. WATOVHD is the only exit benzene has from the process, benzene being made by reaction 3 and consumed nowhere. Keeping the stream open therefore reports the aniline available for recovery while leaving that exit intact. A benzene column is needed before the loop can be closed.

Table 7: Decanter sizing.

Item	Value	Note
Organic flow	13.29 m^3/h	1067.0 kg/m^3
Aqueous flow	17.33 m^3/h	965.8 kg/m^3
Density difference	101.2 kg/m^3	
Design droplet	150 μm	Standard decanter basis
Design settling velocity	8.12×10^{-4} m/s	Slower of the two phases
Diameter	1.25 m	
Length	6.25 m	$L/D = 5$
Residence time	15.0 min	
Droplet settling time	12.8 min	Less than residence time
Interface velocity	4.7×10^{-4} m/s	Below settling velocity

3.4 Vent thermal oxidiser

The vapour distillate from the first column, stream VENT, carries benzene, ammonia and hydrogen. It goes to a thermal oxidiser.

Table 8: Thermal oxidiser sizing.

Item	Value	Note
VENT flow	6.39 kmol/h	277.9 kg/h
Benzene in VENT	2.74 kmol/h	Sets most of the duty
Ammonia in VENT	2.02 kmol/h	
Heat release	2.60 MW	Lower heating value basis
Chamber temperature	870 °C	For $\geq$ 98 % destruction
Combustion and quench air	337 kmol/h	
Chamber diameter	1.6 m	
Chamber length	4.8 m	$L/D = 3$
Residence time	1.07 s	At least 1 s required

The heat release is enough for the chamber to run without support fuel.

3.5 Carbon adsorber on the wastewater

The stripper bottoms still carry a little aniline. A granular activated carbon bed is put on this stream as a polishing step.

The bed volume is set by the carbon holding capacity over a 30-day cycle. The contact time that follows is well above the 10 to 30 minutes usually needed, so the bed is not mass-transfer limited.

The wastewater also carries 4.28 weight per cent cyclohexylamine, which is far more soluble in water than aniline and is not held on carbon. It is taken up with the wastewater treatment in Report 5.

Table 9: Carbon adsorber sizing.

Item	Value	Note
Wastewater flow	4375.8 kg/h	4.41 m^3/h at 40 °C
Aniline in wastewater	216 ppm by weight	0.945 kg/h
Aniline to remove	22.7 kg/day	
Carbon loading	0.2 kg/kg	Aniline on granular carbon
Carbon consumption	113 kg/day	
Cycle length	30 days	
Carbon per bed	3400 kg	7.6 m^3 at 450 kg/m^3
Bed diameter	1.2 m	
Bed height	6.7 m	
Empty bed contact time	104 min	

3.6 Consistency check on the converged flows

All the vessels above were first sized on the Report 3 stream table. The flows changed slightly when the kinetic reactor and the closed recycles were brought in. Every vessel was re-checked on the final converged flows. Table 9 gives the result.

Table 10: Vessel checks repeated on the final converged flows.

Vessel	Criterion	Requirement	Final run
Flash drum	Vapour velocity	≤ 100 % of $u_{\max}$	79 %
Flash drum	Liquid holdup	≥ 5 min	5.27 min
Decanter	Settling vs residence time	Settling shorter	12.8 vs 15.0 min
Oxidiser	Residence time	≥ 1 s	1.07 s
Carbon bed	Cycle length	30 days	30 days

The flash drum gained margin. The vapour rate fell and the vapour density rose in the final run, so the drum now runs at 79 per cent of the flooding velocity instead of 99 per cent. The decanter and the oxidiser moved by less than one per cent and both stay inside their criteria on the original dimensions. No vessel had to be resized.

4 Detailed Design of One Heat Exchanger

The exchanger chosen for detailed design is the condenser on the aniline recovery column, block B3 in the flowsheet, which is the ARCOND service. It condenses the ANRECOV vapour against cooling water. It was chosen because it has the largest duty of any water-cooled service in the plant.

The design was done in Aspen Exchanger Design and Rating in rigorous shell and tube mode, in design (sizing) mode, with the process conditions taken from the converged flowsheet.

Table 11: Process conditions for exchanger B3.

Item	Hot side (shell)	Cold side (tube)
Stream	ANRECOV → ARLIQ	Cooling water
Mass flow	3.400 kg/s	152.78 kg/s
Inlet temperature	97.5 °C	30.0 °C
Outlet temperature	40.0 °C	42.1 °C
Inlet vapour fraction	1.000	0
Outlet vapour fraction	7.2×10^{-5}	0
Inlet pressure	1.0 bar	4.0 bar
Allowable pressure drop	0.1 bar	0.5 bar
Calculated pressure drop	0.055 bar	0.458 bar
Fouling resistance	0.0002 m^2K/W	0.0002 m^2K/W

4.1 Process conditions

The hot fluid is put on the shell side because it is condensing. A condensing stream on the shell side has a large free volume to disengage in and a low pressure drop, and the cooling water goes in the tubes where it can be cleaned mechanically.

Fouling resistance of 0.0002 m^2K/W is applied on both sides, which is the standard value for clean organics and for treated cooling water. This is what sets the area, as Section 4.3 shows.

4.2 Geometry

Table 12: Exchanger B3 geometry.

Item	Value
TEMA type	AES
Shell inside diameter	812.8 mm
Tube outside diameter	19 mm
Tube pitch	25 mm, 30° triangular
Tube length	3.658 m
Number of tubes	721
Number of tube passes	2
Shells in series / parallel	1 / 1
Baffle type	Single segmental, horizontal cut
Number of baffles	4
Baffle spacing	590.6 mm
Material	Carbon steel

TEMA AES is chosen because the removable channel cover on the A head lets the tubes be cleaned without breaking the piping, and the floating head on the S rear lets the bundle move when the tube wall runs 50 °C colder than the shell wall. Carbon steel is acceptable because dry aniline is not corrosive to steel and the water side is treated cooling water.

The baffle cut is horizontal because the shell side is condensing. A horizontal cut makes the vapour flow up and down across the bundle and lets the condensate drain along the bottom of the shell instead of collecting in the baffle windows.

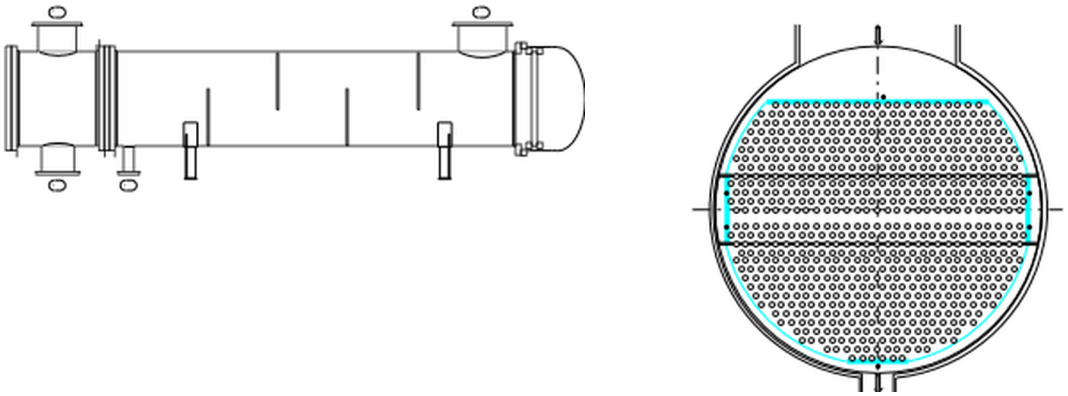


Figure 3: Exchanger B3 from Aspen EDR. Left, setting plan showing the A head, the four single segmental baffles and the S floating head. Right, tube layout of the 721 tubes on 25 mm 30° triangular pitch, with the two tube passes divided by the pass partition.

4.3 Thermal results

Table 13: Thermal results for exchanger B3.

Item	Value
Heat duty	7663.7 kW
Required area	132.87 m^2
Actual area	147.03 m^2
Over-surface	10.7 %
Overall coefficient, clean	2.638 kW/m^2K
Overall coefficient, dirty	1.196 kW/m^2K
UA	158.9 kW/K
Corrected LMTD	48.23 °C
LMTD correction factor	1.819

The over-surface of 10.7 per cent is in the normal range. The dirty coefficient is 45 per cent of the clean one, which shows that fouling, and not the film coefficients, controls the area of this exchanger.

The LMTD correction factor is 1.82, which is greater than one. For a single-phase exchanger this would not be possible, because the correction factor for a 1–2 shell is always below one. Here the shell side is condensing over a wide range, so EDR splits the exchanger into zones and computes a weighted mean temperature difference for each zone separately. The factor of 1.82 is the ratio of that zone-weighted mean difference to the plain end-to-end LMTD. It is above one because most of the duty is released in the desuperheating and condensing zones where the temperature difference is much larger than the plain LMTD suggests.

The duty of 7663.7 kW agrees with the flowsheet value of 7626 kW for the ARCOND service to

within 0.5 per cent, and the cooling water requirement of 550,008 kg/h agrees with the flowsheet value of 549,472 kg/h to within 0.1 per cent.

5 Simulation of the Overall Process with Recycle Loops

5.1 Regressed binary interaction pair

The aniline–nitrobenzene pair decides the separation in the second column, which is the column that sets the final product purity. The Aspen databank parameters for this pair are fitted over 83 to 99 °C only. The second column runs from 129 to 145 °C, so the databank values are being used outside the range they were fitted in.

Report 3 recorded that this pair should be regressed, in Section 10 of that report. This section does that.

Four binary vapour–liquid data sets for aniline and nitrobenzene were taken from the NIST ThermoData Engine inside Aspen Plus and regressed together by maximum likelihood with the Britt–Luecke algorithm and Deming initialisation. Regression case DR-1 in the file `che453_R4_C_regress.apw`. Only the temperature-dependent terms were fitted and the non-randomness parameter was held at 0.3.

Table 14: Binary data sets used in the regression.

Data set	Pressure	Temperature range
BVLE007	0.00667 bar	56–70 °C
BVLE008	0.01333 bar	69–83 °C
BVLE009	0.00416 bar	43–98 °C
BVLE010	Atmospheric	179–210 °C

Taken together the four sets span 43 to 210 °C. The second column runs from 129 to 145 °C, which is inside that span, and BVLE010 brackets it directly. The databank parameters cover only 83 to 99 °C, so they have to be extrapolated to reach the column.

Table 15: NRTL parameters for the aniline–nitrobenzene pair.

Parameter	Databank	Regressed (R-DR-2)
a_{ij}	0	0
a_{ji}	0	0
b_{ij} (K)	−217.928	488.44
b_{ji} (K)	+287.592	−310.108
c_{ij}	0.3	0.3
Valid range	83–99 °C	−273 to 727 °C

Both b terms changed sign. The regressed pair says that aniline and nitrobenzene are closer in volatility than the databank pair suggested, so the split between them is harder than the Report 3 model showed. This is a real result and it changed the design of the second column, as Section 5.3 describes.

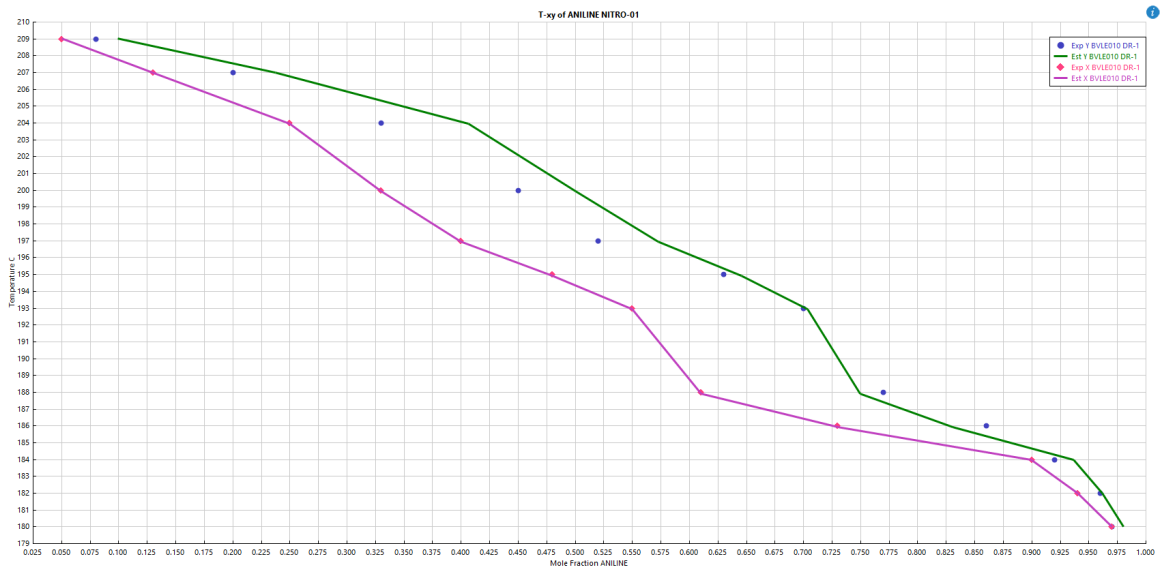


Figure 4: Regressed NRTL fit for the aniline–nitrobenzene pair, data set BVLE010, which is the set closest to the operating range of the second column. Points are experimental, lines are the fitted model.

5.2 Recycle structure and convergence

The flowsheet has four recycle loops. Aspen tears them at the streams listed in Table 14 and converges them with Wegstein. The hydrogen loop is on its own convergence block CV-1 because it carries a design specification on the purge rate. The other three are solved together on $OLVER01.

Table 16: Tear streams and convergence status in the final run.

Tear stream	What it carries	Convergence block	Status
H2RET	Recycle gas to the reactor	CV-1	Converged
S3	Nitrobenzene recycle	$OLVER01	Converged
LIQ-OUT	Flash liquid to the decanter	$OLVER01	Converged
ARLIQ	Aniline recovery liquid	$OLVER01	Converged

All parameter studies were run as separate converged cases rather than with a Sensitivity block, which re-tears every loop at each point when placed on a variable inside a recycle. This is the same practice as Report 3 Section 7.

5.3 Second column: effect of the design variables

With the regressed pair in place, the second column was studied in four converged runs. One variable was changed per run. Table 15 gives the result.

Reflux ratio 2 to 3 took the nitrobenzene from 1896 to 1469 ppm for 36 per cent more reboiler duty. Widening the bottoms draw from 2 to 3 kmol/h took it to 1274 ppm at no extra duty. Report 3 found the same on 15 stages and set both variables aside.

The bottoms composition shows why. At 2 kmol/h the bottoms is already 24 mole per cent nitrobenzene on a 15-stage column, so more reflux only raises the vapour traffic without sharpening the split. The boilup ratio of 249 in run R4_03 is the same evidence.

Table 17: Second column: effect of reflux ratio, bottoms rate and stage count on the product.

Run	Reflux ratio	Bottoms (kmol/h)	Stages	NB in product	Aniline purity
R4_02	2	2	15	1896 ppm	99.29 wt%
R4_03	3	2	15	1469 ppm	99.33 wt%
R4_04	3	3	15	1274 ppm	99.36 wt%
R4_05	3	3	25	**313 ppm**	**99.45 wt%**

Ten more stages took the nitrobenzene to 313 ppm and the aniline to 99.45 weight per cent with no rise in reboiler duty, because the taller column reaches the separation at the same vapour rate. Stage count is the variable that controls this split. The 25-stage column with reflux ratio 3 and a 3 kmol/h bottoms draw is carried forward.

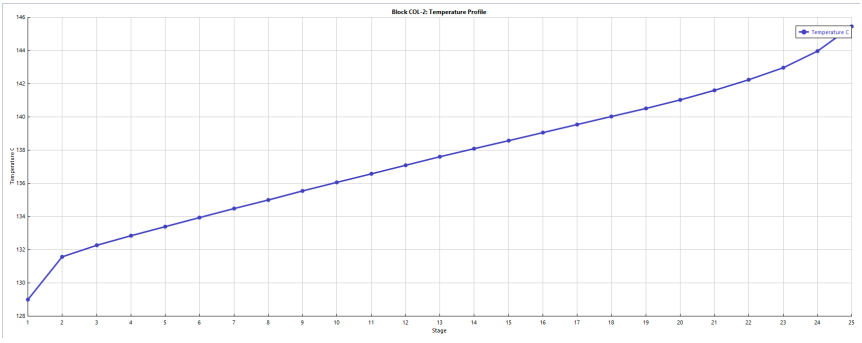


Figure 5: Temperature profile of the second column in the final run.

5.4 Overall mass balance

Table 18: Overall mass balance of the converged flowsheet.

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.16	PRODUCT	131.02	12,185.80
H2-MAKE	444.00	906.59	WASTEWTR	234.35	4,375.76
			WATOVHD	57.48	1,418.94
			VENT	6.39	277.88
			PURGE	7.49	41.49
Total		**18,299.75**	**Total**		**18,299.87**

The balance closes to 0.12 kg/h on 18,300 kg/h, which is 7 parts per million.

5.5 Product

At 8200 operating hours per year this is 99,920 t/y of aniline, which meets the plant basis of 100,000 t/y.

The product leaves at 40 °C and 1.0 bar, and the water content of 0.04 weight per cent is inside the 0.1 weight per cent set in Report 1. Nitrobenzene has come down from 1466 ppm to 313 ppm, a factor of 4.7. The remaining 0.48 weight per cent is cyclohexylamine, taken up in Section 7.

Table 19: Product stream composition.

Component	kmol/h	Mole fraction	Weight
Aniline	130.123	0.99317	99.45 wt%
Cyclohexylamine	0.596	0.00455	0.48 wt%
Water	0.268	0.00205	0.04 wt%
Nitrobenzene	0.031	0.00024	313 ppm
Total	**131.017**		**12,185.8 kg/h**

Table 20: Comparison with Report 3.

Item	Report 3	Report 4 final
Reactor model	Stoichiometric	Kinetic plug flow
Conversion per pass	99.0 % (set)	99.49 % (calculated)
Aniline purity	99.39 wt%	99.45 wt%
Nitrobenzene in product	1466 ppm	313 ppm
Product rate	12,320 kg/h	12,186 kg/h
Second column stages	15	25

5.6 Utilities

The reactor removes a further 18.75 MW, which is not in this table because no utility is assigned to it in the simulation. In practice this heat would raise steam on the shell side of the reactor. That is taken up in Report 5.

Total heating is 22.23 MW against 20.40 MW in Report 3. The difference of 1.78 MW is low pressure steam into the second column, and that is exactly what took the nitrobenzene in the product from 1466 ppm down to 313 ppm.

6 Gas Turbine on the Purge Stream

The purge from the hydrogen recycle loop still carries 6.64 kmol/h of hydrogen. This was run through a gas turbine in a separate file, with natural gas as supplementary fuel and a design specification holding the turbine inlet temperature.

The Report 3 baseline was 709.1 kW of net electrical power. The purge rate changed when the recycle loops were closed, and the turbine gains 5.1 per cent on the new purge.

This case was run on a purge of 7.37 kmol/h taken from an intermediate converged run. The final run gives 7.49 kmol/h, which is 1.6 per cent higher. The turbine output scales almost linearly with the fuel it receives, so the net electrical power on the final purge would be about 757 kW. The turbine is not coupled to the main flowsheet and is reported as a separate case.

7 Objectives Which Could Not Be Accomplished, With Reasons

All four objectives for this report were completed. The items below are carried into Report 5.

Cyclohexylamine in the product. 0.48 wt % against 0.42 wt % in Report 3. It boils below

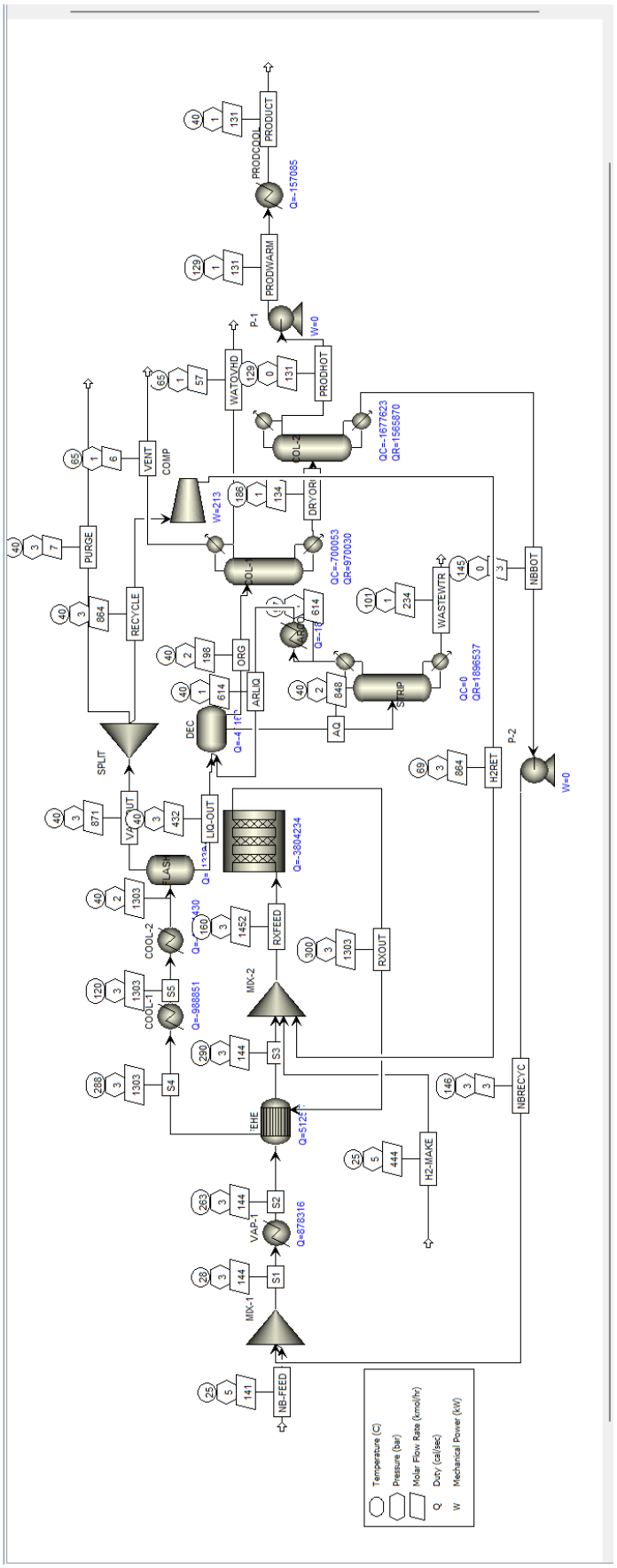


Figure 6: Converged flowsheet with all four recycle loops closed.

Table 21: Utility summary of the converged flowsheet.

Utility	Service	Duty (MW)	Usage (kg/h)
High pressure steam	VAP-1	3.68	8,792
Medium pressure steam	COL-1 reboiler	4.06	7,564
Low pressure steam	STRIP reboiler	7.94	14,199
Low pressure steam	COL-2 reboiler	6.55	11,723
	Total heating	**22.23**	**42,279**
Cooling water	ARCOND	7.63	549,472
Cooling water	COOL-2	5.24	377,467
Cooling water	COL-2 condenser	7.02	506,019
Cooling water	COL-1 condenser	2.93	211,156
Cooling water	PRODCOOL	0.66	47,381
	Total cooling water	**23.47**	**1,691,496**

aniline, so a light-ends draw at the top of COL-2 is the route, as established in Report 3 Section 2.4. This is a change to the column configuration.

Nitrobenzene in the product. 313 ppm against 1466 ppm in Report 3, the target being 5 ppm. Section 5.3 shows the split responds to stage count, so a taller COL-2 takes it further.

Second decanter on WATOVHD. Reported open loop in Section 3.3, because WATOVHD is the only exit benzene has. A benzene column is needed before that loop can be closed.

Reactor heat. 18.75 MW is available from REACT as a steam-raising duty. Its assignment belongs with the heat integration left to Report 5 in Report 2.

8 Any Other Challenges

Reacting phase had to be set to Vapor on all three reactions. On Liquid every rate came out as zero in an all-vapour stream. A Reset was needed after the change, the tear streams otherwise holding the earlier solution.

The reactor profile grid needed 41 points. On the default grid the peak read 353 °C and 589 °C against 391 and 640 °C on the fine grid, the integrator stepping over the hot spot.

Ergun was replaced by a fixed pressure drop inside the recycle loop. The correlation is used to fix the value at the design flow, as Section 2.3 describes, and the converged answer is unchanged.

Design specifications were used in place of sensitivity analyses, as in Report 3, each parameter case being run as its own converged file. One change was made per run and each run saved as a new numbered file, `che453_R4_00` through `che453_R4_05`.

9 Number of Hours Spent During This Period

Approximately 8 hours per member, which is 48 hours for the group.

Table 22: Gas turbine on the purge stream.

Item	Value
Purge to turbine	7.37 kmol/h
Hydrogen in purge	6.52 kmol/h
Supplementary natural gas	7.71 kmol/h
Air	290 kmol/h at 12 bar
Turbine inlet temperature	1153.9 °C
Turbine exhaust temperature	593.3 °C
Air compressor work	864.9 kW
Fuel compressor work	16.3 kW
Turbine work	1657.3 kW
Net shaft power	776.1 kW
Net electrical power	745.1 kW

10 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the kinetic model and its rate parameters, the conversion of the reactor to RPlug, the bed geometry and pressure drop, convergence of the flowsheet with all four recycles, the COL-2 study, and the compilation of this report.

Nithin D H and Shubham Singh — the flash drum, decanter, second decanter, thermal oxidiser and carbon adsorber, the re-check of every vessel on the converged flows, and the gas turbine case on the new purge.

Shivam Dongare and Sajag Masane — the regression of the ANILINE–NITRO-01 pair against NIST TDE data, and the detailed design of exchanger B3 in Aspen EDR with its fouling and over-surface checks.

All six members have reviewed the complete report.

11 References

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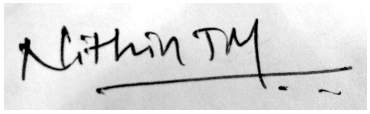
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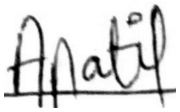
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14. Own regression case R-DR-1, ANILINE–WATER NRTL parameters, Report 1 Section 6.4.
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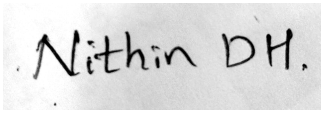
Signatures of members



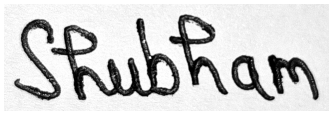
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Aditya Nitin Patil



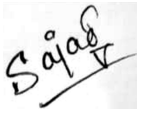
Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane